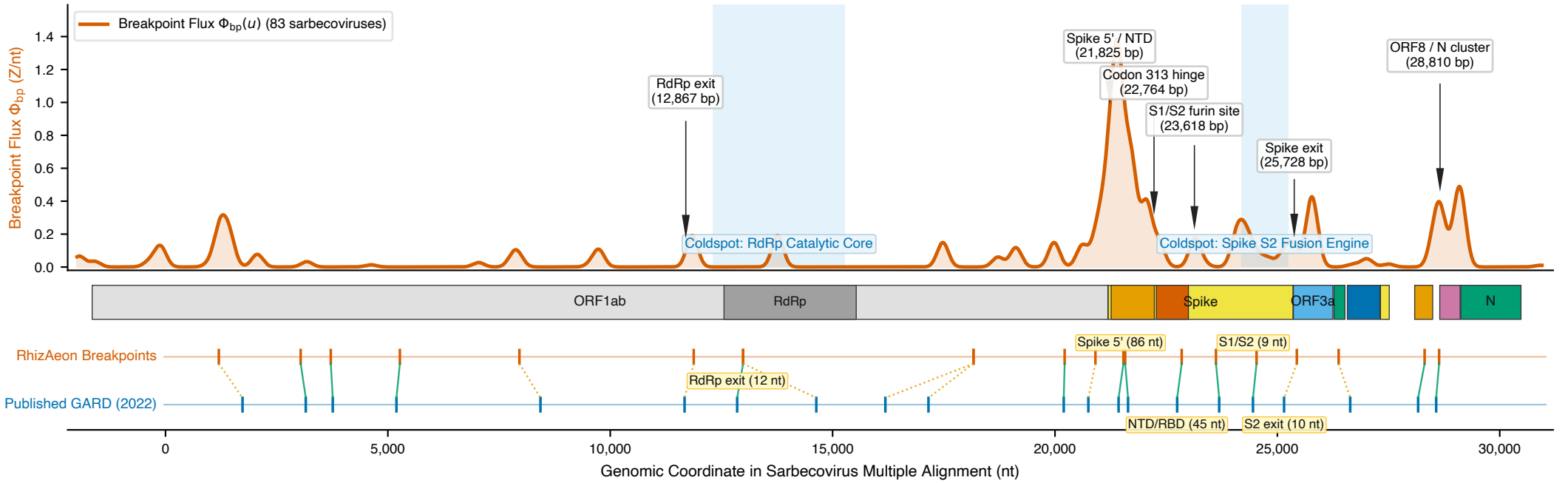
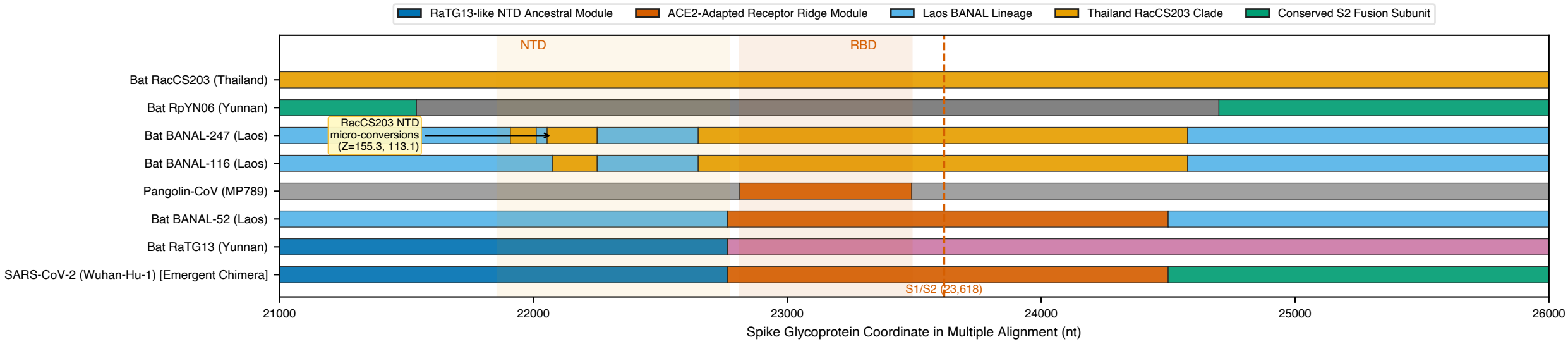
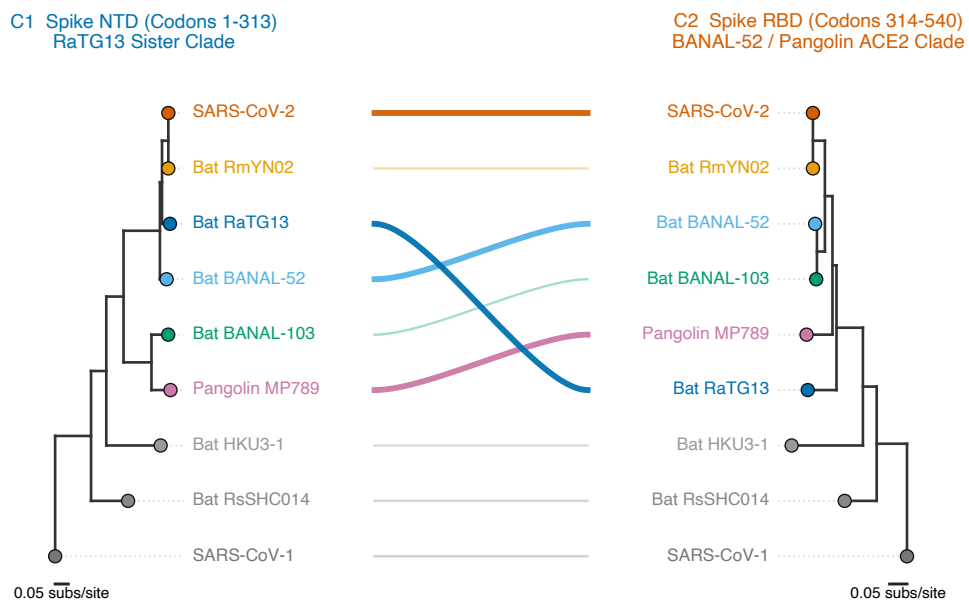
A Genome-wide Recombination Flux $\Phi_{bp}(u)$ and Benchmark Concordance Across 30.9 kb



B Spike Reticulate Mosaic Architecture Across Wildlife Progenitors and Emergent SARS-CoV-2



C SARS-CoV-2 Incongruence Across Codon 313 Hinge



D Polarized Ancestral Recombination Graph (ARG) for Sarbecovirus Progenitors

